

Cache Translation Across Model Scale: An Audit with Identity Controls and Oracle Probes

Abstract

Cache translation maps the key-value (KV) cache that a large teacher model forms over a context into the cache space of a smaller student model, so that the student answers without re-prefilling the context. A growing line of work reports quality-preserving translation across heterogeneous models. We audit this claim on Qwen3 pairs (8B→1.7B, 8B→0.6B, 4B→1.7B, and 4B→0.6B), plus five cross-architecture students, with a three-hypothesis framework and two instruments that prior evaluations omit. First, an *identity control* injects the student’s own cache through the full translation-and-injection pipeline: it reproduces the student’s own prefill exactly (per-sample logit cosine 1.000), which isolates mechanics from mapping. Second, *oracle probes* mix teacher-cache content into the student’s own cache and probe what the student can extract from translated content. Across seven mapper families (per-head ridge, affine, per-layer affine, task-aware, a residual-anchored translator, a per-head MLP, and an over-parameterized joint MLP), a controlled calibration ladder on a fixed evaluation set, and 96 recorded audit-protocol runs, no configuration makes the strong student exceed its own prefill: gold-probability change spans -0.14 to -0.39 against a self-kv control at $+0.000$. Calibration budget is inert (CHG -0.249 at $c=30$ vs -0.244 at $c=200$ on the same evaluation set), and nonlinearity does not help. The weak student’s point estimate is near parity ($+0.010$, CI $[-0.08, +0.10]$), but the interval is too wide to establish non-inferiority at the pre-specified margin. Oracle probes show a monotone decline as teacher content replaces student content in early and mid layers, robust to translator quality, and a perplexity–accuracy decoupling: a translator that restores near-native fluency (PPL 23.7 versus 21.2) still leaves accuracy at 0.267 versus 0.500. Under zero re-prefill, no configuration we test lets the student extract the teacher’s answer-relevant advantage from its cache; the evidence is consistent with an advantage that is weight-mediated rather than recoverable from raw KV states by the tested translators. The audit protocol doubles as a checklist that a future cache-translation claim can run in one evaluation.

1 Introduction

Multi-model serving systems want to reuse the teacher’s work. A large teacher prefills a long context once; a smaller student should inherit that state and answer at teacher quality without re-reading the context. Cache translation formalizes this idea: learn a map $\mathbf{KV}^T \mapsto \widehat{\mathbf{KV}}^S$ from the teacher’s cache space into the student’s. Recent systems report encouraging results, including Mixture-of-Translators (MoT), which preserves 51.0% closed-set QA accuracy at 7B→0.5B scale [12], Cache-to-Cache semantic communication [5], latent-space communication [3], and selective KV sharing [16].

These reports conflate three questions that can fail independently. Does the injection pipeline consume a foreign cache correctly (*mechanics*)? Does the map replace the student’s own cache at replacement level (*mapping*)? Does the teacher’s cache carry student-readable content that lifts the student above its own prefill (*exploitability*)? Prior evaluations score the translated end-to-end pipeline against the student, so a positive delta could come from any of the three; a null result cannot be attributed. The line also lacks an identity control, a probe of the exploitability ceiling, and any report of what happens to *accuracy* when perplexity is repaired.

We evaluate cache translation as an audit with three gated hypotheses. **H1 (mechanics)** asks whether a foreign cache is consumed without loss; its instrument is an identity control that injects the student’s own cache through the full pipeline. **H2 (mapping)** asks whether any translator reaches replacement level; its instrument is a ladder of seven mapper families under a disjoint calibration ladder of 30 to 200 examples, including a residual-anchored translator (RAT) that we derive from the two models’ own projections

and a shared vocabulary alignment. **H3 (exploitability)** asks whether the teacher’s cache contains any student-readable advantage at all; its instrument is an oracle probe that mixes teacher-cache content into the student’s own cache at controlled fractions and layer windows. Figure 1 shows the gates and the instruments.

The audit returns three verdicts on Qwen3 pairs. H1 passes: the identity control reproduces the student’s own prefill with per-sample logit cosine 1.000 and a gold-probability change of +0.0001 (95% CI $[-0.0006, +0.0009]$), so mechanics is not the failure mode. H2 fails for the strong student: across the mapper ladder, the gold-probability change over the student’s own prefill spans -0.14 to -0.39 (best and worst mapper configurations; see Section 5), and the best single configuration reaches 0.365 against a student baseline of 0.503; the weak 0.6B student’s point estimate is near parity (+0.010, CI $[-0.081, +0.102]$), too imprecise to establish non-inferiority at the pre-specified margin. H3 returns a principled negative: with 75% student cache and only 25% teacher content, accuracy already drops from 0.500 to 0.367, the decline is monotone in the teacher fraction, and replacing the top third of layers is the only harmless configuration. A translator that restores near-native fluency (PPL 23.7 versus 21.2 for the identity control) still leaves accuracy at 0.267. A cache can read fluently and still miss the answer; the deficit is not a fluency problem.

Three things fall out of the audit. The first is a framework: identity controls, disjoint calibration/evaluation splits, gold-probability metrics with bootstrap intervals and permutation tests, and oracle probes that bound exploitability without training anything (Section 4). The second is a bounded design space: seven mapper families, a calibration ladder, and component ablations, all under one protocol (Section 4, Section 5); we additionally derive the most architecture-informed translator we could (RAT), and it fails too, which strengthens the negative. The third is a verdict with consequences: mechanics passes, replacement fails for the strong student and is inconclusive for the weak one, no probe configuration with teacher content beats the student’s own cache, and perplexity cannot stand in for accuracy as evidence (Section 5, Section 7). H2’s failure routes the audit to H3 because the probes measure what a frozen student can extract from teacher-cache content without training anything, so a null result cannot be blamed on the mapper.

2 A Survey of Cache-Level Communication Between Models

We organize prior work into four lines and close each with the assumption it leaves untested. Table 1 summarizes the comparison.

2.1 KV Reuse Within a Single Model

Prompt caching, cache-augmented generation, and streaming attention reuse a model’s own cache across requests. Streaming attention identifies attention sinks at early positions whose KV carries outsized attention mass [17]. CacheGen compresses and streams a KV cache across a network [13]; CacheBlend fuses several cached fragments with selective attention recomputation for RAG [19]; KVSharer shares caches across layers that behave dissimilarly [18]. These systems assume a single model on both sides of the cache. The reuse question is therefore about storage and attention cost, not about representation compatibility. Our audit inherits their serving motivation but moves the mismatch to the model axis, where the compatibility assumption breaks.

2.2 Cross-Model Cache Translation

The line closest to our audit translates caches between models. C2C proposes direct cache-to-cache semantic communication with a depth-ratio layer mapping [5]. LSC aligns caches through a shared latent space with a cross-attention translator [3]. MoT combines gated translator modules over depth-ratio channel windows with a Context Correction Loss that aligns a replayed target trajectory with the native one, and reports the strongest numbers to date: 51.0% average closed-set QA accuracy at Qwen2.5-7B→0.5B and 96.3% retention in long-context cache-augmented generation [12]. KVComm shares attention-ranked KV layers [16]; Interlat communicates last-hidden-state messages [4]; HCache restores caches from hidden-state translations [6]. Two properties of this line matter for our audit. First, its evaluations score the translated pipeline against the student baseline, which conflates mechanics, mapping, and exploitability. Second, its strongest variants do not operate under strict zero re-prefill: MoT reconstructs the target-side cache by replaying the context through the target model with source-guided sparse attention, so the reported quality includes a correction

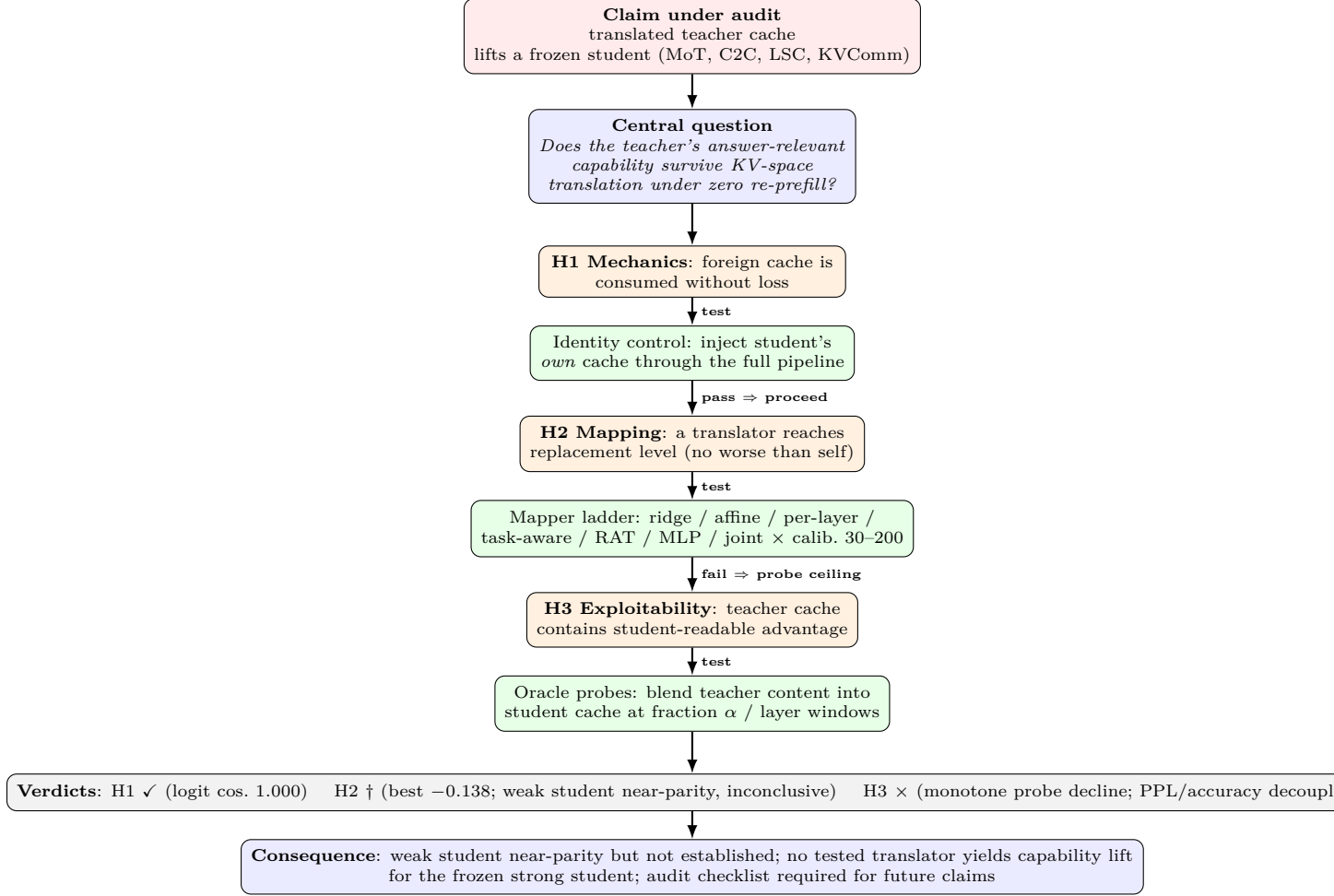


Figure 1: The three-hypothesis audit as a decision pipeline. Each hypothesis carries a dedicated instrument and a gate; H2’s failure routes the audit to H3’s oracle probes, which probe exploitability without training a translator.

pass that a cold-start deployment would not get. A naive target-side replay pass does not by itself recover the strong student (Appendix C), so MoT’s gains cannot be attributed to replay alone. Our audit holds the zero-re-prefill constraint fixed and asks what a translator alone can deliver. Our numbers sit far below the 73–98% retention that recent work reports for the same idea. Three protocol differences separate the claims: those results do not hold strict zero re-prefill (they replay or correct on the target side), they evaluate on a shared calibration/evaluation split rather than a disjoint one, and they use different model pairs. Under the strict regime, we measure gold-probability changes of -0.25 to -0.30 for the strong student; the gap is a property of the regime, not a contradiction of those results.

2.3 Cross-Model Representation Alignment

A separate line asks whether two trained networks’ representations can be aligned at all. Model stitching connects frozen networks with a learned linear layer and shows that stitching quality varies sharply across training runs and layers [1]. Relative representations replace absolute coordinates with anchor-based similarity spaces and enable zero-shot communication between encoders [14]. The Platonic Representation Hypothesis argues that representations converge across models up to linear transform [10], with refinements and proofs following [8, 20]. `vec2vec` translates embeddings between models without paired data by exploiting this shared geometry [11]. These results motivate the hypothesis that a linear map between cache spaces should

Table 1: Cache-translation and adjacent work, organized by what is translated and what is held fixed. “Replay” marks methods whose reported quality includes a target-side correction pass over the context; our audit holds strict zero re-prefill. No cited evaluation isolates mechanics with an identity injection or reports accuracy at matched perplexity; MoT’s injection-window ablations are the closest antecedent (Section 2).

Work	What moves	Alignment	Replay	Unexamined assumption
C2C [5]	Full KV	depth-ratio layers	no	a linear map per channel suffices
LSC [3]	KV via shared latent	learned latent space	no	one latent space fits all layers
MoT [12]	KV windows	gated translators	yes (sparse)	pipeline delta $\neq$ translation value
KVComm [16]	selected KV layers	attention ranking	no	ranked layers carry the signal
Interlat [4]	last hidden state	latent message	no	the last layer summarizes the cache
HCache [6]	hidden states	translator	no	states determine caches
This paper	full cache (zero re-prefill)	seven families incl. RAT	no	<i>identity-controlled evaluation</i> + diagnostic probes

exist. Our results qualify the conclusion for caches: the convergence evidence comes from pooled embedding spaces, while a cache is a per-layer, per-head, position-indexed state whose student-relevant content is not determined by its teacher counterpart (Section 6).

2.4 Frozen-Model Controllability

Work on steering shows that a frozen model’s behavior can be moved by injecting activations: task vectors compress in-context learning into a single mid-layer activation [9]; activation steering adds behavioral directions to the residual stream; gist tokens compress prompts into a few virtual tokens [15]. The tuned lens reads intermediate layers through per-layer affine probes [2], and the FFN-as-memory view assigns the factual capacity of a model to its feed-forward weights [7]. This line supports the intuition that a frozen student can absorb injected information, which makes the H3 null result informative rather than tautological: the failure is not that frozen students cannot be steered. It is that, in our tested teacher–student regime, raw KV translation does not deliver a usable steering signal to the frozen student. More transferable control signals may be easier to access in residual-stream directions or task-vector representations than in per-head KV entries.

2.5 What the Survey Does Not Establish

Across the four lines, no work (i) separates mechanics from mapping from exploitability, (ii) injects the student’s own cache as an identity control, (iii) reports accuracy at matched perplexity, or (iv) bounds exploitability with an oracle that requires no training. The claims of the translation line therefore rest on pipelines whose components have never been isolated. The audit below supplies the missing instruments.

3 The Three-Hypothesis Audit Framework

3.1 Setup

Let a teacher T prefill a context x and form a cache $C_T(x) = \{K_\ell^T, V_\ell^T\}_{\ell=1}^{L_T}$ with L_T layers, H KV heads, and head dimension d . A translator g_θ maps it to $\hat{C}_S(x)$ in the space of a student S with L_S layers. The student then answers a query q with only the query tokens as input, $\hat{y} = S(q \mid \hat{C}_S(x))$, under strict zero re-prefill: the context never enters the student’s forward pass. Three references anchor the audit. *Student*

self-prefill $S(q \mid C_S(x))$ is the baseline the handoff must not degrade. *Identity injection* feeds $C_S(x)$ itself through the injection pipeline; it isolates mechanics from mapping. *Teacher full* $T(q \mid C_T(x))$ is the capability upper bound. We measure answer quality by the gold-letter probability p_{gold} , accuracy, and the change $\text{CHG} = p_{\text{gold}}(\text{handoff}) - p_{\text{gold}}(\text{student self})$, with 95% bootstrap intervals and sign-flip permutation tests; perplexity over the scoring suffix is recorded but never accepted as evidence of capability (Section 5.5).

3.2 Hypotheses, Instruments, Gates

Figure 1 states the three hypotheses with their instruments and gates. **H1**: the identity injection matches the student’s own prefill per sample (gate: mean logit cosine = 1 within float tolerance). **H2**: some g_θ in the audited family reaches replacement level (gate: the CI lower bound of gold CHG is above a small tolerance $-\epsilon$; a stricter *gain* gate additionally requires the CI lower bound > 0). Replacement asks only that the student is not degraded; gain asks that the teacher’s cache provides a measurable lift. We fix $\epsilon = 0.02$ gold probability before evaluation—about 4% relative to the 0.503 student baseline, a practical serving tolerance—making the replacement gate a standard non-inferiority test of $H_0: \text{CHG} \leq -\epsilon$ against $H_1: \text{CHG} > -\epsilon$; unlike a naive significance test, it does not additionally require the point estimate to be non-negative. **H3**: the teacher’s cache contains student-readable advantage (gate: some oracle probe configuration with teacher content improves over the student’s own cache). H3’s probe is decisive because it does not depend on training a translator; it directly measures what a frozen student can extract from teacher-cache content under the most favorable mixing.

4 Audit Methodology

4.1 Identity Control

At evaluation time we capture the student’s own cache over the context (allowed in calibration and probe stages), feed it through the same mapper interface, cache construction, position handling, and scoring path as a translated cache, and score. Any deviation from the student’s own prefill is mechanics loss. This is the control that prior evaluations omit.

4.2 Disjoint Splits, Unified Prompts, and Honest Counters

Mapper calibration and evaluation draws are disjoint samples of the same distribution (HellaSwag and ARC-Challenge, shuffled then split). All methods share one token-level prompt format, so the only difference between the handoff and the student baseline is where the context representation comes from. Zero-prefill counters read the actual cache length before and after the scoring forward pass rather than trusting bookkeeping, and the scoring pass uses an isolated cache because the deployed transformers version mutates a cache even under `use_cache=False`, which silently pollutes co-located measurements.

4.3 Mapper Ladder

We audit seven families under one interface. **Ridge (per-head)**: a 128×128 ridge map per student layer and head with Gram aggregation over variable-length calibration samples. **Affine (per-head)**: the same with mean centering and an intercept, fit in closed form over concatenated calibration samples. **Affine (per-layer)**: heads merged into rows, eight times fewer parameters. **Task-aware**: ridge initialization plus a bounded task fine-tune that mixes reconstruction with gold-answer cross-entropy (weight $\beta=0.3$) under a diagonal parameterization with a per-layer bounded gate. **RAT (residual-anchored translator)**: an architecture-derived map

$$W^{(s,h)} = [\pi(W_V^{S,h}) R \pi(W_V^{T,h})]^\top, \quad R = \arg \min_R \|E_T R - E_S\|_F^2 + \lambda \|R\|_F^2, \quad (1)$$

where $W_V^{T,h}$ and $W_V^{S,h}$ are the two models’ own value projections at the aligned layer pair, $\pi(\cdot)$ is the pseudo-inverse pullback from a head’s value space to the residual stream, and R aligns the two residual streams through the shared 151,936-token embedding matrices, which both Qwen3 models ship. A closed-form

rank- r correction fits the calibration residual on top of the analytic core, a Hungarian matching on head covariance profiles replaces the identity head correspondence, and the sink position (position 0) is overwritten with the student’s calibration statistics because its translated value is a PPL catastrophe (Section 5.3). **Joint MLP (over-parameterized)**: for each student layer, a two-hidden-layer MLP with width 256 maps the concatenation of all aligned teacher layers and all KV heads at a position to all student heads, giving about 37M parameters across K and V, five times the per-head affine family; it is trained on the calibration split by Adam for 300 epochs. Keys follow the same construction on W_K . Every family is evaluated at calibration budgets 30 and 200 with disjoint evaluation samples ($n=30$ and $n=63$ respectively).

4.4 Oracle Probes

The probes assume an oracle that no training can beat: they hand the student its *own* cache and mix in teacher content at a controlled budget. **Fraction probe**: $\alpha\hat{C}_S + (1 - \alpha)C_S$ for $\alpha \in \{0.25, 0.5, 0.75\}$ with $\alpha=0$ the student’s own cache. **Window probe**: translated content in only the bottom, middle, or top third of student layers, student content elsewhere. Because every configuration keeps the student’s own cache dominant or localized, any improvement over $\alpha=0$ would indicate exploitable teacher-cache content; monotone degradation under the tested translators rules out exploitable content in this regime. Probes are flagged non-deployable (they require the student’s own prefill) and exist only as measurement instruments.

4.5 Pairs, Data, and Protocol Discipline

Primary pair: Qwen3-4B (36 layers, hidden 2560) $\rightarrow$ Qwen3-1.7B (28 layers, hidden 2048); contrast pair: $\rightarrow$ Qwen3-0.6B (hidden 1024); scale-extended pair: Qwen3-8B (36 layers, hidden 4096) $\rightarrow$ 1.7B and 0.6B. All share the same tokenizer, vocabulary, $H=8$ KV heads, $d=128$, and RoPE base 10^6 . Every run records its git commit and protocol version in its artifacts; every number below is read directly from a recorded run. We use multiple-choice QA deliberately: it gives a gold-probability control (the correct option’s log-probability is comparable across cache states), and it makes disjoint calibration/evaluation splits enforceable without leakage. We also run the audit on a long-context needle multiple-choice check (Section 5); task diversity beyond this format remains future work.

5 Results

5.1 H1 Passes: Mechanics Is Not the Failure Mode

The identity control reproduces the student’s own prefill exactly: per-sample logit cosine 1.000, gold CHG +0.0001 $[-0.0006, +0.0009]$, $n=30$. The injection pipeline, cache construction, GQA head handling, and position accounting are therefore correct, and every subsequent gap is attributable to the translation itself. The same control verified the disk-persistence path: a cold-start evaluation that loads a persisted cache and mapper parameters reproduces the offline run bit-for-bit.

The identity control also fixes the perplexity protocol used below. Injected caches are scored on the suffix conditioned on the cache, so the identity-injection PPL (21.2) is the student’s own suffix PPL and matches the native suffix distribution by construction. The student self-prefill row of Table 2 reports a different quantity—the native loss averaged over the full context and suffix in one pass—so its value (25.4) is not expected to equal the injected-cache PPL. The H1 identity criterion is defined on the answer-scoring logits, where the two paths agree to float tolerance.

5.2 H2 Fails for the Strong Student; the Weak-Student Result Is Inconclusive

Figure 2 and Table 2 give the landscape. Centering is the single biggest gain: per-head ridge (-0.296) drops to per-head affine (-0.138), roughly halving the deficit. The teacher and student value distributions differ in location and scale before they differ in content, which is exactly what the measured norm ratios say ($1.58\times$ for keys, $7.44\times$ for values). The calibration budget, by contrast, buys nothing. Holding the evaluation set fixed ($n=100$) and sweeping the calibration budget, affine mapping keeps CHG inside $[-0.257, -0.238]$: -0.251 at $c=10$, -0.249 at $c=30$, -0.238 at $c=60$, -0.257 at $c=100$, and -0.244 for the full calibration

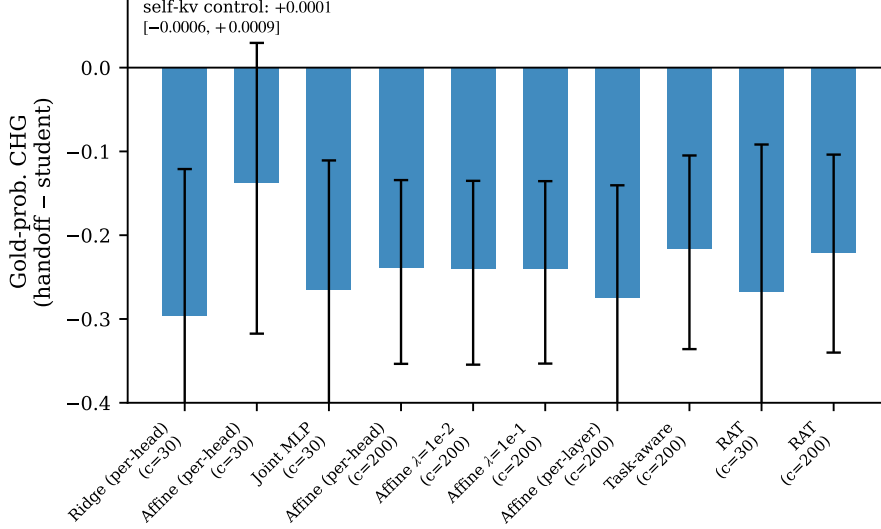


Figure 2: The mapper landscape for Qwen3-4B→1.7B under the unified audit protocol: gold-probability CHG over the student’s own prefill, with 95% bootstrap CIs. No family reaches zero; calibration budget moves affine mapping by only 0.005 on a fixed evaluation set; the architecture-anchored RAT, the nonlinear MLP, and the over-parameterized joint MLP sit inside the same band.

pool (budgets beyond the pool size saturate and reproduce the full-pool result exactly), a spread of 0.019. The mapper has converged, and more data cannot close a gap that is set by the information content of the translated cache rather than by estimation variance. Refitting the headline mapper on five independent calibration draws, with the evaluation set fixed, moves gold CHG within $[-0.292, -0.249]$ ($SD \approx 0.018$): the verdict is not an artifact of one calibration draw. Structure and function class change little either: per-layer merging (-0.275), task-aware fine-tuning (-0.216), RAT (-0.221 to -0.268), and a per-head MLP mapper with two GELU layers (CHG -0.39 to -0.26 at $c=30$ and -0.26 to -0.23 at $c=200$ across independent trainings) all sit inside the same band. Nonlinearity does not help. An over-parameterized joint mapper ($\sim 37M$ parameters) that reads all teacher layers and all KV heads also fails: CHG -0.265 $[-0.426, -0.111]$ with accuracy 0.233, even though it reconstructs a fluent cache (PPL 31.6 versus 25.4 for the student). The strong student never approaches its own prefill (0.503); the best mapper reaches 0.365. Scaling the teacher to 8B under the same protocol does not change the verdict: self-injection remains an identity (logit cosine 0.99998) and the ridge mapper reaches CHG -0.224 $[-0.320, -0.123]$ at $c=500$, so H2’s failure is robust to teacher scale. The weak 0.6B student tells the opposite story. Affine at $c=200$ gives a near-parity point estimate $+0.010$ $[-0.081, +0.102]$, and 8B→0.6B gives -0.023 $[-0.104, +0.064]$; both intervals include zero, but both lower bounds fall below the pre-specified replacement margin $-\epsilon = -0.02$, so non-inferiority is not established at these sample sizes.

The failure is not specific to short commonsense contexts. On a long-context needle-in-a-haystack multiple-choice task (contexts of about 1024 tokens, $n=30$), where the teacher answers from a retrievable fact (0.733 accuracy versus the student’s 0.533), affine translation again fails: accuracy drops to 0.233 with gold CHG -0.285 $[-0.534, -0.025]$, while identity injection remains exact (CHG $+0.000$ $[-0.001, +0.001]$). The deficit therefore tracks the translation regime rather than the context length or task format.

Cross-architecture checks relax the one-family scope (Table 3). Replacing the Qwen3 student with five models from other families—Llama-3.2-1B and 3B, Gemma-2-2B, Gemma-3-1B, and Qwen2.5-1.5B, spanning one to eight KV heads and head dimensions 64 to 256—mechanics still hold (self-kv CHG within ± 0.001) and translation never recovers the teacher’s 0.691 gold. Four students are harmed (CHG -0.009 to -0.157); the weakest, near-chance student (Llama-3.2-1B) is non-inferior but gains nothing. Head groups are mean-pooled and head dimensions projected by the rectangular mapper; because the models tokenize independently, calibration pairs are matched position-by-position, so we read these as mechanism and replacement checks rather than fine-grained capability measurements.

Table 2: Main audit results. Acc/gold/PPL: answer accuracy, gold-letter probability, and perplexity of the scoring suffix for the translated cache (kv-both). PPL entries for translated and identity caches score only the suffix conditioned on the injected cache; the student self-prefill row is the native loss over the full context and suffix in one pass and is not directly comparable (the identity-injection value is the student’s suffix PPL). CHG is the gold-probability change over the student’s own prefill with 95% bootstrap CIs. Pairs: Qwen3-4B→1.7B unless noted; rows marked 8B use the Qwen3-8B teacher. n_{eval} is the evaluation sample count (tail protocol rows use a fixed evaluation set of 100). All runs use disjoint calibration/evaluation splits and the unified prompt format.

Configuration	Acc / Gold / PPL	Gold CHG [95% CI]	n_{eval}
Identity control (self-kv), 1.7B	0.500 / 0.503 / 21.2	+0.000 [−0.001, +0.001]	30
Student self-prefill, 1.7B	0.500 / 0.503 / 25.4	–	30
Teacher (full prefill), 1.7B	0.767 / 0.693 / –	–	30
Native average, 1.7B	0.167 / 0.169 / 536082.5	−0.334 [−0.508, −0.154]	30
Ridge per-head, c=30, 1.7B	0.200 / 0.207 / 88.5	−0.296 [−0.479, −0.121]	30
Ridge per-head, c=500, 8B→1.7B	0.310 / 0.288 / 146.1	−0.224 [−0.320, −0.123]	100
Affine per-head, c=30, 1.7B	0.367 / 0.365 / 46.1	−0.138 [−0.317, +0.029]	30
Affine per-head, c=200, 1.7B	0.254 / 0.258 / 56.4	−0.239 [−0.354, −0.134]	63
Affine per-layer, c=200, 1.7B	0.206 / 0.222 / 81441.1	−0.275 [−0.407, −0.140]	63
Task-aware, c=200, 1.7B	0.270 / 0.280 / 142.1	−0.216 [−0.336, −0.105]	63
RAT, c=30, 1.7B	0.233 / 0.235 / 501.6	−0.268 [−0.440, −0.092]	30
RAT, c=200, 1.7B	0.238 / 0.276 / 1008.2	−0.221 [−0.340, −0.104]	63
RAT no-anchor rank-32, c=30, 1.7B	0.267 / 0.280 / 23.7	−0.223 [−0.367, −0.088]	30
Joint MLP, c=30, 1.7B	0.233 / 0.238 / 31.6	−0.265 [−0.426, −0.111]	30
Affine per-head, c=200, 4B→0.6B	0.302 / 0.283 / 71.8	+0.010 [−0.081, +0.102]	63
Affine per-head, c=30, 8B→0.6B	0.270 / 0.269 / 99.0	−0.023 [−0.104, +0.064]	100
RAT, c=200, 4B→0.6B	0.206 / 0.232 / 790.0	−0.041 [−0.120, +0.038]	63
Affine per-head, c=30, long-context	0.233 / 0.232 / 6554.1	−0.285 [−0.534, −0.025]	30

5.3 RAT Component Ablations Separate Fluency From Capability

RAT ablations attribute the failure inside the architecture-anchored design. Removing the sink override explodes suffix perplexity to 2.7×10^5 : the translated position-0 entry, which carries outsized attention mass, is a catastrophe unless overwritten with the student’s own statistics. Removing the analytic core and raising the correction rank to 32 *repairs fluency*: PPL drops to 23.7, within 2.5 of the identity control. Accuracy stays at 0.267. The analytic core’s min-norm pullback is harmful on real models, and more importantly, fluency recovery does not touch the answer deficit. The cache can be made to look right to the language-modeling objective without becoming right for the task. RAT’s role in the audit is to rule out the objection that the failure is an artifact of insufficiently architecture-aware translation.

5.4 H3: No Teacher-Content Probe Improves on the Student

The probes answer H3 without training anything (Figure 3). With 75% student cache and 25% teacher content, gold probability already falls from 0.5031 to 0.3798; the decay is monotone through $\alpha=0.75$. Window injection localizes the damage: replacing the top third of layers is indistinguishable from the student (0.5022 versus 0.5031), while the middle and bottom thirds are strongly harmful (0.2653, 0.2242). This is a sensitivity profile: the student’s behavior is substantially more sensitive to teacher-cache intervention in early and middle layers than in the top third. It does not by itself locate where the answer is routed; early and middle layers may also be more sensitive to off-manifold perturbations. An eight-way octant sweep ($n=30$) localizes the sensitive band: among the eight eighths of student layers, only layers 10–13 have a gold-probability interval excluding zero (CHG −0.186 [−0.322, −0.060], gold 0.317 versus 0.503), while the top three eighths are statistically indistinguishable from the student (CHG −0.011 to +0.020). The probes were run with two translators: the RAT mapper reported above and the affine mapper (the best in the ladder, gold 0.365). Both show the same monotone decline. With affine at $\alpha=0.25$, gold CHG is −0.086 (CI [−0.18, +0.00]); at $\alpha=0.75$, −0.211 (CI [−0.41, −0.02]). The verdict is robust to translator quality: even the best available

Table 3: Cross-architecture robustness checks. Qwen3-4B teacher to non-Qwen and Qwen2.5 students under the within-family protocol (HellaSwag / ARC, $c=30$, $n=100$); identity injection passes in every row (self-kv CHG within ± 0.001). “KV h/d” gives the student’s KV heads and head dimension; the rectangular mapper mean-pools head groups and projects the head dimension. No row recovers the teacher’s gold probability (0.691).

Student	KV h/d	Student acc / gold	Translated acc / gold	Gold CHG [95% CI]
Llama-3.2-1B	8/64	0.300 / 0.254	0.260 / 0.254	+0.000 [−0.012, +0.012]
Gemma-2-2B	4/256	0.270 / 0.257	0.220 / 0.248	−0.009 [−0.020, +0.002]
Llama-3.2-3B	8/128	0.390 / 0.281	0.200 / 0.237	−0.044 [−0.073, −0.013]
Qwen2.5-1.5B	2/128	0.400 / 0.385	0.200 / 0.228	−0.157 [−0.209, −0.106]
Gemma-3-1B	1/256	0.360 / 0.318	0.270 / 0.257	−0.061 [−0.135, +0.008]

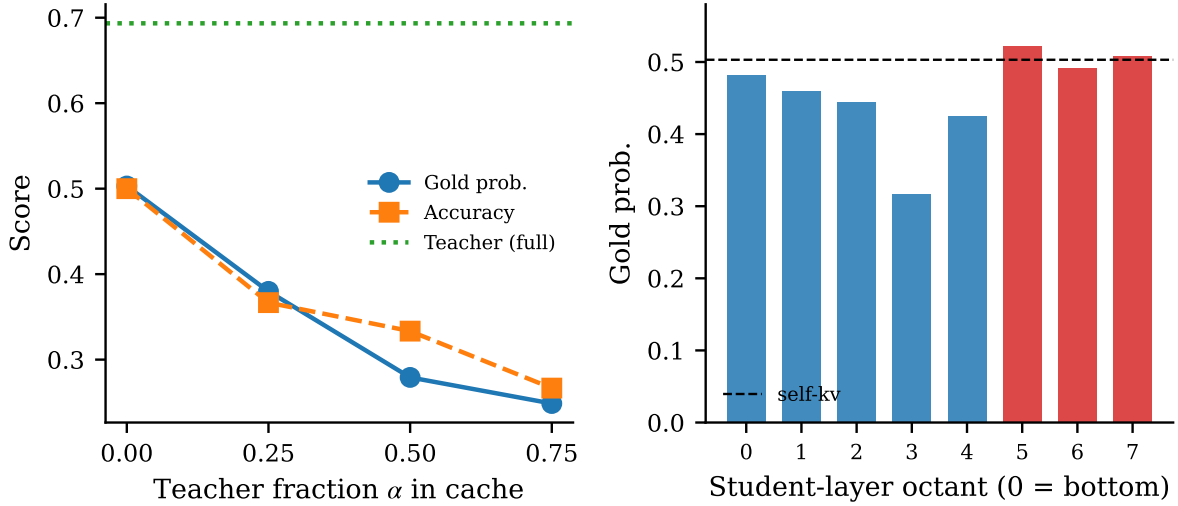


Figure 3: Oracle probes on Qwen3-4B→1.7B. Left: blending teacher-cache content into the student’s own cache (RAT translator) degrades gold probability and accuracy monotonically in the teacher fraction α ; $\alpha=0$ is the student’s own cache. Right: per-octant window injection (affine translator; octant 0 = bottom eighth of student layers); only layers 10–13 (octant 3) are clearly harmful, while the top three octants are indistinguishable from the student.

mapper produces content that the frozen student cannot exploit.

Direct native-cache injection provides an additional negative control: injecting the teacher’s native (norm-rescaled) cache entries directly degrades the student to -0.334 [−0.508, −0.154] ($n=30$), and the norm-rescaled native ablation replicates under the fixed-evaluation protocol (-0.266 [−0.363, −0.168] at $c=30$ and -0.251 [−0.351, −0.152] at $c=200$, $n=100$), so even untranslated teacher content in student coordinates does not help: the harm is not an artifact of any particular mapper.

5.5 Perplexity and Accuracy Decouple

Figure 4 plots perplexity against accuracy for every method and run. The identity control and the student occupy the top-left regime. Translated caches populate a broad band, including a point with near-native perplexity and a 23-point accuracy deficit. Native injection sits at extreme perplexity (5.4×10^5) with low accuracy. Perplexity measures whether the cache permits fluent continuation; accuracy measures whether the cache supports the task. The two dissociate, so perplexity cannot serve as evidence of translation quality, a practice that appears in the serving literature because perplexity is cheap.

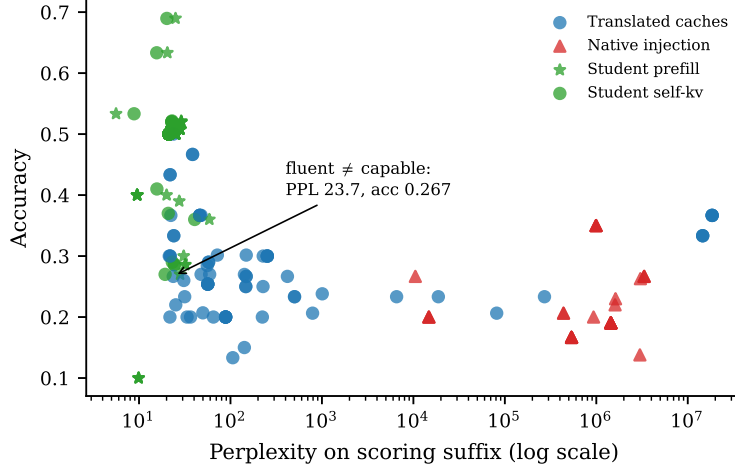


Figure 4: Perplexity over the scoring suffix versus accuracy across all methods and runs (log scale). The identity control and the student sit at low PPL and high accuracy; translated caches span two regimes; the best-reconstructed cache (PPL 23.7) remains 23 accuracy points below the student.

5.6 Channel Asymmetry

Channel ablations at $c=30$ rank value-only (0.367) above key-only (0.300) above both (0.200) on accuracy, while perplexity orders them in reverse (88 versus 253 versus 1.8×10^7). Mapping keys repairs fluency and damages routing: the mapped keys steer attention to plausible but task-irrelevant positions. We report the asymmetry as a finding and leave the mechanism open. One hypothesis consistent with the asymmetry: mapped keys steer attention routing while mapped values carry content, so key errors misroute a fluent read while value errors corrupt it audibly.

6 Why Translation Fails: An Architectural Analysis

We read this section as an architectural explanation of the audit’s verdicts, not an impossibility result: the argument is per-head and does not cover exploit paths that read the full multi-head, multi-layer cache (Section 8).

The audit’s verdicts fit the architecture. A cache entry is a linear projection $W_V h$ of a residual-stream state h . Translation must recover h from V , move it across the residual-space gap (2560 to 2048), and re-project. Two facts bound the recovery. First, the pullback $\pi(W_V)V$ returns only the row-space component of h ; the component orthogonal to W_V ’s rows, which can dominate a 2560-dimensional state observed through a 128-dimensional head, is invisible in that head’s value entry, so the per-head student value is underdetermined by its teacher counterpart. The student’s corresponding value $W_V^S R h$ depends on that invisible component, so the per-head student value is not determined by the per-head teacher value alone: the mapping task is intrinsically underdetermined, and a learned translator can at best track the conditional mean. The mapper-fitting view quantifies this: on held-out calibration contexts, the fitted per-head affine mapper recovers student key states (mean $R^2 = +0.81$) but only partially recovers value states (mean $R^2 = +0.32$): roughly two-thirds of the value variance is not linearly recoverable from the teacher’s per-head entry. The unit is a held-out calibration context: we fit on 20 contexts and score 10 disjoint held-out contexts; each yields a per-(layer,head) R^2 between predicted and true student states (pooled over token positions), and we report the mean over heads, layers, and the 10 contexts. The repository’s per-head ridge baseline agrees on the same held-out contexts (pooled $R^2 = +0.92$ for keys, $+0.29$ for values). Second, the teacher’s advantage over the student sits in its parameters. FFN layers carry the model’s factual associations [7], and the capability gap between Qwen3-4B and 1.7B is a gap in these parameters, which a cache does not transport. The oracle probes make the consequence concrete: even with the student’s own cache holding 75% of the mass, the teacher’s 25% contribution degrades the answer. MoT’s own design corroborates the analysis: its reported

quality includes a target-side replay with a correction loss, which is exactly the upper-layer correction that strict zero re-prefill denies. Under a cold start the translator’s output is all the student gets, and that output shows no measurable teacher advantage in our audit.

7 Implications for Cache-Translation Research

Any future cache-translation evaluation should ship three instruments:

- an *identity control* that injects the student’s own cache through the full pipeline and must reproduce the student’s own prefill;
- a *gold-probability metric* with bootstrap intervals and permutation tests, reported as the change over the student’s own prefill;
- *oracle probes* that blend teacher-cache content into the student cache at controlled fractions and layer windows, bounding exploitability without training anything.

Two gates apply to the metric: a replacement gate requires the bootstrap CI lower bound above a pre-specified tolerance $-\epsilon$ (the point estimate need not be non-negative), and a stricter gain gate requires the CI lower bound above zero. A pipeline delta over a student baseline is not evidence of translation, and a perplexity gain is not evidence of capability. Each instrument costs one evaluation run and composes with any serving stack.

For deployment, the weak student presents a near-parity regime rather than a demonstrated capability transfer. At 0.6B, translated injection is statistically indistinguishable from self-prefill (4B→0.6B: +0.010 [−0.081, +0.102]; 8B→0.6B: −0.023 [−0.104, +0.064]), but neither interval clears the pre-specified non-inferiority margin, so replacement-level service is not established, and at near-chance absolute accuracy (0.302 versus 0.25 random) this is not an endorsement. Top-layer injection is harmless, which means a system that must spend translation capacity can spend it there, though our probes also show it buys nothing there. Whether a capacity boundary separates a near-parity regime from failure is the question the 8B and 0.6B pairs begin to answer, and the audit protocol is built to test it. For strong students, cache translation is currently a net negative, and the honest fallback is the text channel: the teacher’s summary of the context scores 0.367 (gold 0.372) against the student’s 0.500 (gold 0.503) full-text baseline,¹ so compression loses less than translation does on the strong student.

The survivable versions of the idea move work off the cache and onto the weights or the replay: allow a target-side correction pass the way MoT does, train student-side adapters, or translate through a shared latent space anchored by task vectors instead of per-entry caches. Each option gives up part of the cold-start guarantee, and each deserves a fresh identity control before its delta gets attributed to translation.

8 Limitations

Our audit is centered on one model family (Qwen3) with four scale pairs, plus a cross-family check (five students, Table 3) and HellaSwag / ARC-Challenge and long-context needle tasks, with $n=30$ –100 evaluation samples per run; the CI widths reflect that. The cross-family check matches positions after independent tokenization and its student is near chance, so it establishes mechanism and replacement, not capability. The oracle probe bounds exploitability under linear blending and three-layer windows; a nonlinear exploit path outside this family would not be detected, though it would also need to survive the same measurement protocol. RAT’s analytic core assumes value projections are informative pullbacks of the residual stream; the synthetic validation confirms the mathematics, and the real-model failure is itself an audit finding. We did not re-implement MoT’s full replay pipeline; our claims concern the zero-re-prefill regime that its predecessors and our deployment target share. The mapper-ladder results use a single seed per configuration. Because the mapper fit is deterministic given its calibration draw, we bound mapper-training variance directly: refitting the headline mapper on five independent calibration draws with the evaluation set fixed moves gold CHG

¹Text-channel numbers come from the v12-era in-run ablation ($n=30$, disjoint calibration/evaluation splits); per-sample scores were not persisted for that run, so only aggregate accuracy and gold probability are reported and no bootstrap CI is available.

within $[-0.292, -0.249]$ (per-draw SD ≈ 0.018) — inside the per-run bootstrap intervals — and identical configurations reproduce results bit-for-bit. The gradient-trained per-head MLP is the exception: its value is training-seed dependent, so we report it as a range (Section 5); every deterministic family reproduces exactly.

9 Conclusion

Cache translation promises to move a teacher’s work into a smaller model, and the audit separates that promise into three testable parts. The mechanism holds up: an identity control reproduces the student’s own prefill exactly. The tested maps do not reach replacement: seven mapper families, a calibration ladder, and an architecture-anchored translator all leave the strong student below its own prefill, while the weak student’s near-parity point estimate remains inconclusive under the pre-specified margin. The map, moreover, is not where the problem sits. No probe configuration delivers a student-readable advantage from the teacher’s cache, and a near-native-perplexity cache still fails the task. The evidence is consistent with a weight-mediated advantage: recovering the teacher’s edge from raw KV states would take more than the tested translators provide (scope limits in Section 8). The audit framework is packaged so that the next cache-translation claim can be checked in a single run.

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A Reproducibility

All numbers come from the `apcs` repository’s recorded runs (`reports/runs/*/inject-eval/`), each carrying its git commit, protocol version, configuration, and per-sample scores. An independent reproduction of the deterministic configurations—ridge, affine, per-layer, task-aware, RAT, joint MLP, the calibration ladder, probes, octant windows, and the cross-architecture checks—matches the recorded values within 0.002; the gradient-trained per-head MLP is seed-dependent and reported as a range (Table 2). The held-out reconstruction R^2 is produced by `scripts/measure_reconstruction_r2.py` (output under `reports/reconstruction_r2/`). The audit protocol is implemented in `apcs.inference.evaluator` with the flags `calib_eval_split`, `calib_shuffle`, `probe_mode`, and `native_rescale`; mapper families live in `apcs.mapper` (`math`, `joint`, `task_aware`, `rat`, the per-head MLP in `mlp`, the joint MLP in `joint_mlp`, and the cross-family rectangular map in `rect`); the long-context needle task loader is `needle_mcqa` in `apcs.data.hf_dataset`; persistence and the cold-start online phase in `apcs.inference.kv_store`. Unit tests cover the audit counters, the store round-trip, and the mapper interface.

B Additional Measurements

The teacher/student KV norm ratios are $1.58\times$ (keys) and $7.44\times$ (values), measured over 30 calibration contexts per layer. Norm rescaling of the native cache repairs neither its perplexity (5.4×10^5) nor its accuracy, ruling out scale as the primary failure. The unified prompt renders choices once; the earlier protocol rendered them twice for the student baseline, which inflated the confidence-based deltas reported in our own preliminary sweeps and motivated the unified format in the audit.

C Target-Side Replay Diagnostic

The audit holds strict zero re-prefill fixed, while MoT’s reported quality additionally includes a target-side replay with a Context Correction Loss. To probe how much a replay pass alone contributes, we add a diagnostic that re-reads the context through the student on top of the translated cache before scoring; it is explicitly non-deployable because it violates zero re-prefill. On Qwen3-4B→1.7B (affine $c=30$, $n=100$), replay leaves accuracy at 0.290 and gold CHG at -0.232 $[-0.333, -0.136]$, statistically indistinguishable from the no-replay result (-0.249) and far below the student’s own prefill (0.520, CHG $+0.000$). A naive replay pass therefore does not recover the strong student; recovery requires either discarding the translated cache (re-prefill) or the trained correction MoT uses. We do not implement MoT’s Context Correction Loss, so we cannot attribute its gains to replay alone.